AUTO INSURANCE CLAIMS OPTIMIZATION EXECUTIVE BUSINESS REPORT

Prepared for Executive Leadership, Risk & Underwriting Teams

Executive Summary

Rising claim frequency and severity are reducing profitability. This report demonstrates how analytics can identify high-risk policyholders, uncover claim drivers, and support risk-based business actions. The proposed approach combines exploratory analysis, predictive modeling, and operational recommendations aimed at reducing costs and improving underwriting outcomes.

1. Business Problem Statement

The insurer is experiencing an increase in claims that is negatively affecting loss ratios and overall profitability. Traditional underwriting approaches may not adequately distinguish high-risk customers from lower-risk customers. The business objective is to identify policyholders with elevated claim risk, improve pricing accuracy, and enable proactive risk mitigation.

2. Project Objectives and Success Metrics

a. Success will be measured using:
b. Number of high-claiming individuals identified,
c. Approximate cost savings generated from actions, and
d. Identification of claim drivers that improve underwriting and risk decisions. Example targets include a 10% reduction in claim costs and 8% improvement in profitability.

3. Data Summary

a. Example dataset contains policy id , customer demographics, vehicle information, claim history, and policy tenure.
b. Analytical dataset contains 58,592 policy records and 2 years of historical claim information.
c. There are 15 numeric columns and 11 categorical columns.
d. There are some outliers in the data. After removing Outliers the shape of the data is 54480 rows and 44 columns
e. The imbalance ratio of the target variable(is_claim) is: 15:1. The data is highly imbalanced.

Data Quality and Preprocessing

Exploring the data set with checking first five rows of dataset, shape of data set, explore the data type of columns like float, integer, string, looking for missing values, looking for duplicate, checking for number of unique values in all columns and dropping unnecessary columns.

Key Findings

1. The dataset has no missing values
2. No duplicate found in dataset
3. Policy no are unique, the dataset contains 22 area cluster, 5 car make (Brand), 11 models of car, 3 fuel type.
4. Age of car and policy holder is normalized.

5. Exploratory Data Analysis (EDA)

1. EDA revealed that a relatively small group of customers contributes a disproportionate share of claim costs. Claim concentration was also observed in selected area cluster.
2. Key findings
3. Total sales of policies are 58592
4. Higher the policy tenure your more of the claims
5. Average claim rate of the policies is 6.4%.
6. Total claims are 3748
7. The policies are mostly sold in C8 area cluster.
8. C18 has the highest claim rate of 0.11
9. Make 1 and 2 has most claim rate
10. Model M2 has highest claim rate.
11. The possibility of claim decreases with increase in the age of the vehicle
12. New cars have more claims
13. High displacement (CC) vehicle has 5 more claims than lower one.
14. Policy holders normalise age is above 0.8 has higher claim rate

6. Feature dropping

Dropped feature that highly correlated above 0.9.

1. Displacement
2. Turning radius
3. max_power
4. length
5. max_torque
6. width

Dropping feature with high VIF

1. gross_weight
2. Height
3. Power_to_weight_ratio

4. ncap_rating
5. airbags'

7. Feature Engineering Findings

Key engineered features include High risk driver flag, safety score and Power to weight ratio. These features improved predictive performance and business interpretability.

8. Model Development and Evaluation

Two candidate models were evaluated: Logistic Regression and Decision Tree. Model comparison on metrics like Accuracy, Precisions, Recall, F1 score and roc-auc.

Best model – Decision tree classifier
Evaluation metrics = roc-auc
Best Parameters:

Max depth	5
Min samples split	17
Min samples leaf	9
Criterion	Entropy

8. Business Implications and Recommendations

Adopt risk-based pricing, strengthen underwriting requirements for high-risk applicants, implement proactive monitoring of claim-heavy customer segments, and enhance fraud detection review processes. Use model outputs to prioritize intervention activities.

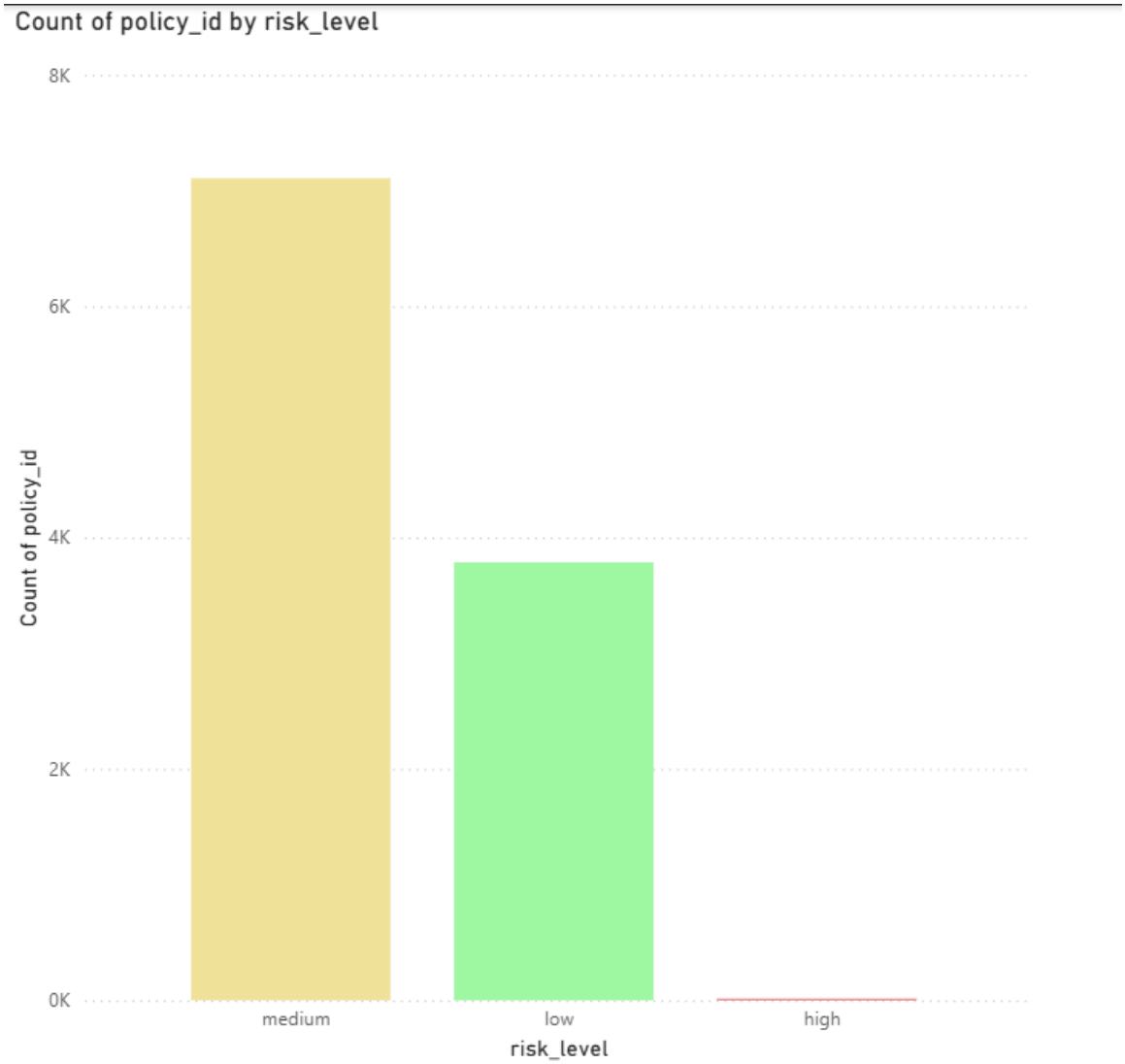
9. Limitations and Next Steps

This report uses illustrative examples. Future enhancements could leverage telematics data, weather information, repair network data, and real-time behavioral signals. Regular model monitoring and recalibration are recommended.

Illustrative KPI Dashboard

Metric	Result	Business Value
High-Risk Customers	3	Targeted monitoring
Model Accuracy	52.3%	Improved segmentation

Figure 1: Claim Concentration by Risk Segment



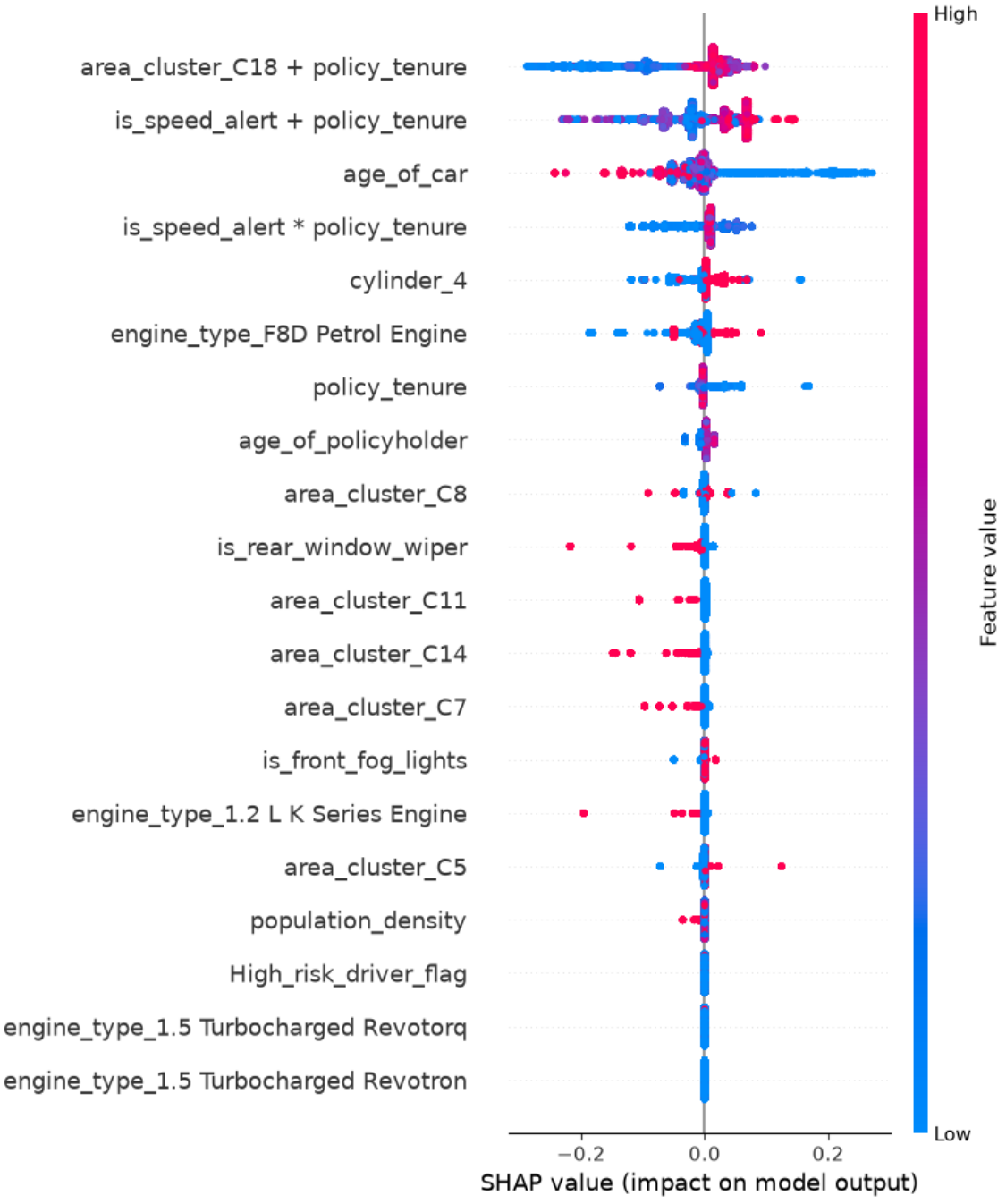
Business insight: High-risk policyholders contribute disproportionately to claim volume and should be the focus of pricing and monitoring strategies.

Figure 2: Key Claim Drivers

Prior claims and annual mileage are illustrative leading indicators of future claim risk.

1. area_cluster_C18 + policy tenure
2. is_speed_alert + policy tenure
3. age_of_car
4. is_speed_alert * policy_tenure

5. cylinder_4
6. engine_type_F8D Petrol Engine
7. policy_tenure
8. age_of_policyholder
9. area_cluster_C8
10. is_rear_window_wiper



Model Selection Rationale

Decision tree provides the strongest predictive performance while preserving explainability through feature importance analysis. It therefore represents the recommended production model.

Executive Recommendations

Implement risk-tier segmentation, automate underwriting checks, and establish monthly monitoring dashboards for high-risk segments. Expected benefits include improved premium adequacy, stronger risk selection, and lower claim costs.